

Single-mode tapered optical fiber for temperature sensor based on multimode interference

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Abstract

A temperature sensor using a single-mode tapered fiber coated by thermo-sensitive material is presented. It works on the multimode interference influenced by the small change of the ambient refractive index. To better understand the tapered optical fiber, simulations that change parameters such as the taper waist diameter and the ambient refractive index are performed using RSOF BeamPROP. It is illustrated that optical losses vary with the ambient refractive index, and reduce diameter within a certain range can increase the sensitivity. In our experiment, with the high thermo-sensitive coefficient of material, a good temperature sensing result was achieved. The range of temperature measured is from -20°C to 80°C . The results show that the temperature sensor has high temperature sensitivity and good repeatability.

Keywords: temperature sensor, tapered optical fiber, thermo-sensitive material

1. INTRODUCTION

Temperature is of significant for both the scientific and industrial fields. In recent years, a number of studies based on fiber Fabry-Perot interferometer (F-P) [1], fiber Bragg gratings (FBGs) [2], Raman scattering [3], cladding-mode resonant special fiber sensor [4], optical fiber coupler [5] etc have been proposed to realize it. Though most of these sensing systems have good performance, they suffer from some drawbacks. F-P requires complex fabrication methods. Sensors based on Raman scattering require expensive detection system. FBGs require precise and expensive phase masks and stringent photolithographic procedures. Cladding-mode resonant special fiber sensors require special fiber. Sensors based on tapered optical fiber have recently been report as an excellent candidate for sensing due to its low cost and easy. Some of these sensors include chemical sensor based on evanescent field absorption [6], tapered fiber-optic surface plasmon resonance hydrogen sensor [7], tapered optical fiber humidity sensor with nanostructured sensitive coating [8], nanoresonator temperature sensor [9], strain sensor made from tapered holey optical fiber [10], RI sensor based on Mach-Zehnder made from two taper interferometer [11], and so on.

In this paper, we present a temperature sensor based on tapered optical fiber coated by thermo-sensitive material. Since the material has a high thermo-optical coefficient, the changed refractive index of the coating film with the ambient temperature variation influence the interference of the multimode, and finally leads to variation of the fundamental mode output. Compared to existing techniques, for instance FBGs, the tapered optical fiber has the advantages of simple demodulation, mature tapered fabrication techniques, high sensitivity and low cost.

2. THEORETICAL ANALYSIS

Taper fiber is normally made by pulling the optical fiber while it is heated to reduce the diameter to tens of micrometers. See Fig.1, the tapered fiber consists of three parts: a waist region with small and uniform diameter, and two taper regions with gradually changed diameter beside the waist region, see Fig.1. In the first untapered region, which is single mode fiber, all light are guided by the core, and the only transverse mode is fundamental mode HE_{11} . According to the wave theory, the normalized frequency directly determines the properties of light in the core, given as:

$$V = \frac{\pi d}{\lambda} \sqrt{n_{cor}^2 - n_{cla}^2} \quad (1)$$

However, at the first taper region, when the core diameter decreases to a certain value, resulting in the normalized frequency V being approximately equal to unity, the light cannot be confined in the core and spreads to the cladding. Thus, the cladding acts as new core, and the material surrounding is now the new cladding. As the new core diameter increases and the larger NA due to the larger refractive difference between the cladding and the air, resulting in the new V , higher order modes can be supported and a multimode interference behavior occurs. At the second taper region, these modes gradually couple back to the HE_{11} mode of the single-mode optical fiber. Mathematical description of such structure has been presented by Ref.[12]. They assumed the region of core mode cutoff is treated in the sudden approximation, and the whole taper region satisfies Stewart-Love adiabatic criterion [13], so that there is only a beating between excited local modes. The measured output intensity is described by:

$$I = \sum_{m=1}^N I_m + 2 \sum_{m>n} \sum_{n=1}^N \sqrt{I_m I_n} \cos \phi_{mn} \quad (2)$$

with

$$\phi_{mn} = \int_0^L (\beta_m(z) - \beta_n(z)) dz \quad (3)$$

where N is the total number of modes supported by multimode region, I_m is the intensity of HE_{1m} mode, L is length of multimode region, $\beta_m(z)$ is the propagation constant of HE_{1m} mode which is depended on the ambient refractive index.

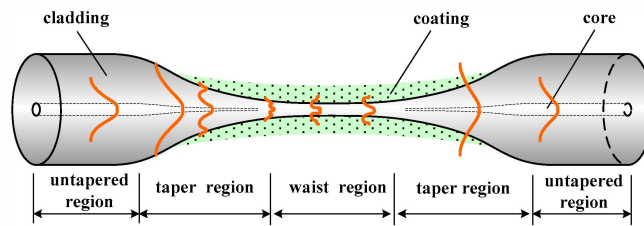


Fig.1. Scheme of the proposed temperature sensor structure

We use a commercially available software package RSOF BeamPROP to analyze the output intensity of the tapered optical fiber with the different ambient refractive index. In the simulation, the exponential profile in tapered region is adapted with the taper length of 8000 μ m. The core diameter and cladding diameter of single mode fiber is 8.8 μ m and 125 μ m respectively, and the uniformly waist region length is 4000 μ m with the diameter of 10 μ m, 15 μ m, and 20 μ m. The index of core and cladding are assumed 1.46 and 1.457, respectively. And the ambient refractive index is taken from 1.36

to 1.44. The wavelength is 1550nm.

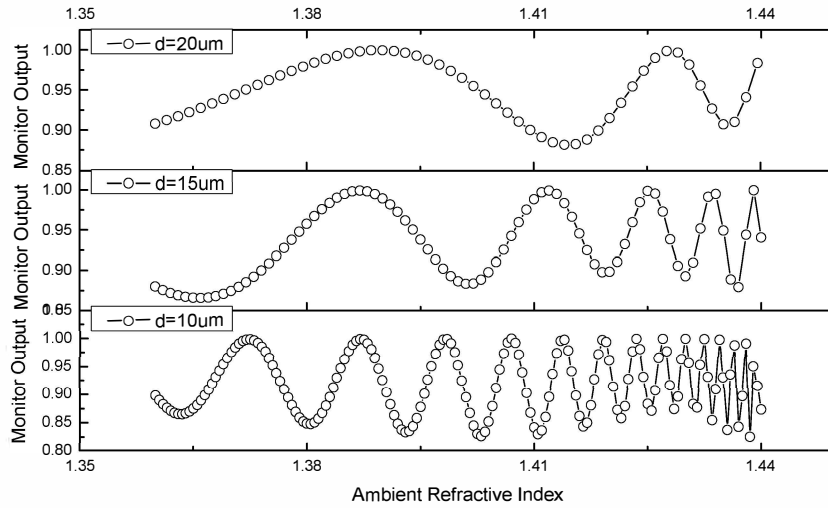


Fig.2. Relationship of the normalized output intensity and the ambient refractive index

In Fig.2, the output intensities oscillate with the ambient refractive index, and the period of the oscillation increase with decreasing of the ambient refractive index. If the output intensity is a monotonic function of ambient refractive index in a certain measuring range, its sensitivity to the refractive index can be used for sensing application. It is also noticeable that the period of the output intensity changes decreases along with the decreasing of waist diameter, which means that high sensitivity can be obtained by decreasing the waist diameter and the large measuring range by increasing that.

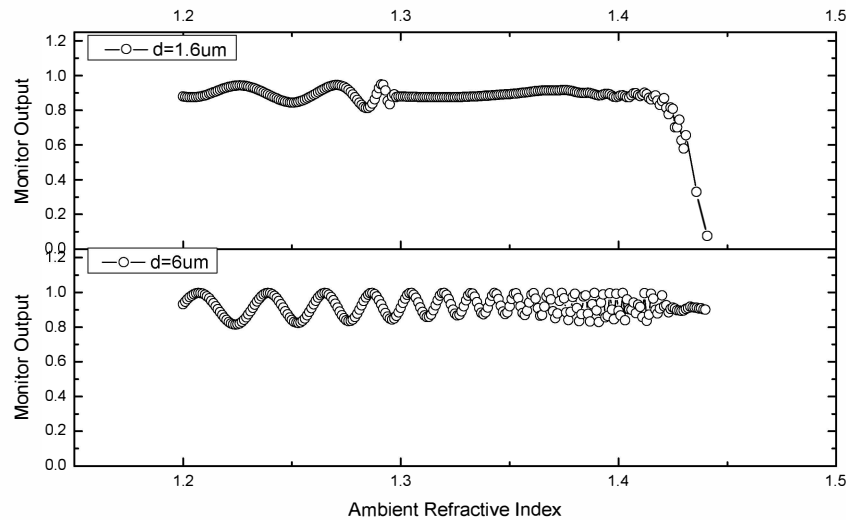


Fig.3. Output intensity for the taper waist diameters of 1.6 μ m (upper portion) and 6 μ m (lower portion)

Further, we change the taper waist diameter to 1.6 μ m and 6 μ m, and the ambient refractive index is taken from 1.2 to 1.44. See Fig.3, the oscillation of the output intensities remains when the taper waist diameter is 6 μ m. However, when the taper waist diameter is 1.6 μ m, the output intensities oscillate first, then flatten and finally decrease sharply along with the ambient refractive index increase. It can be explained as follow. If the diameter of the taper waist is less than a certain value, the interference will not occur for 1550nm light with a certain ambient refractive index, since the waist of such

size is similar to the core of single mode which is no longer support higher modes. When the diameter of the taper waist continues to decrease or the ambient refractive index changes until the normalized frequency V is approximately equal to a unit, the taper waist can not guide light any longer. So the taper waist needs to be controlled in order to ensure the multimode interference. The diameter can be calculated using Equation (1) by introducing $V = 2.405$, the cladding refractive index and the ambient refractive index. Fig.4 shows the relationship of the normalized frequency V and the taper waist diameter in the case of different ambient refractive index.

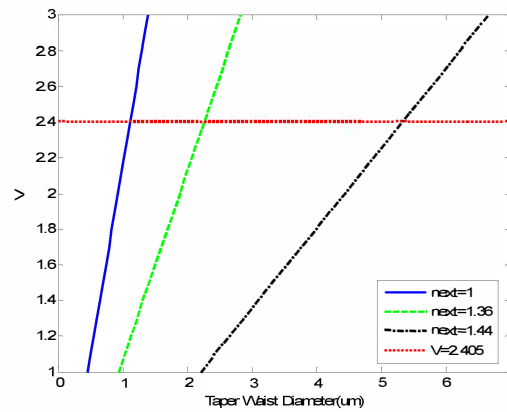


Fig.4. Relationship of V and taper waist diameter under different ambient refractive index

3. EXPERIMENT AND RESULTS

In this work, we used heat-pulling system with oxyhydrogen flame to fabricate the tapered optical fiber. The single-mode fiber was used. Its protective coating was removed about 1cm, and then stretched by two motors while it was heated by flame. The motors moved to the opposite side until it reached to the define position. The fiber must be keep straight during the fabrication processing. As is discussed above, since the material has a higher thermo-optical coefficient than pure silica, temperature variation about 100 degrees can result in a larger ambient refractive index changes. In order to make the output intensity change monotonically over this range of the ambient refractive index, we control the taper waist diameter and total taper length to be about 20μm and 20mm according to the simulation results. The insertion losses of these tapers are less than -2dB. These parameters depend on the heater range, the pulling length and speed which can be controlled by computer. The taper was packaged in a quartz groove, and coated by thermo-optical material. The size of the sensor head is about 2cm, and the thickness of the coating is about 0.5mm.

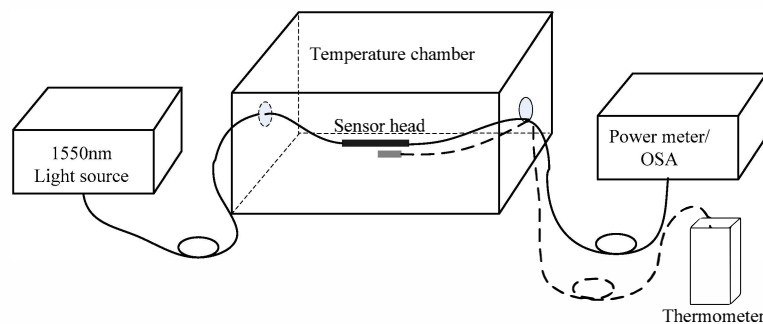


Fig.5. Experimental set-up to characterize the sensor

In the temperature sensing experiment, the sensor head was placed into a temperature controlled chamber (ESL-04KA). Light from a 1550nm SLED light source was launched into the sensor head, and the transmitted light was collected by an optical spectrum analyzer (AQ-6315A), and the values of output power were monitored an optical power meter (AV6334B). To record the real temperature around the sensor head, a commercial thermometer was attached near the sensor head. The experimental set-up is shown in Fig.5.

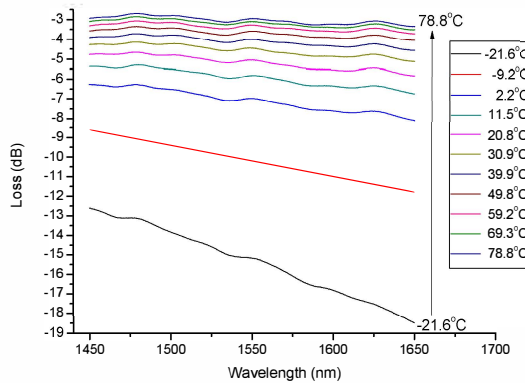


Fig.6. Transmission spectrum by change with temperature

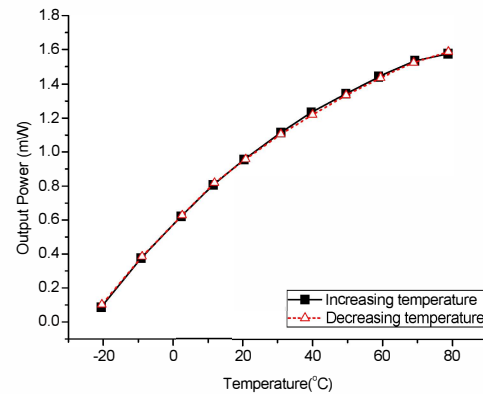


Fig.7. Relationship of output power and temperature

Fig.6 shows the transmission increases with increasing temperature from -20°C to 80°C . As can be seen, the transmission losses gradually decrease during the thermal cycle, therefore more optical power transmitted at higher temperature. For this range of temperatures the output intensities change magnetically in accord with the expected performance of the sensor. This measurement can be conducted for the single temperature detection. Moreover, since the refractive index of the thermo-optic material is a monotonic function of the temperature, the output intensities change magnetically over this range of ambient refractive index, which agree with the numerical simulation.

In order to investigate repeatability of temperature response, we first increased the temperature from -20°C to 80°C , and then performed decreasing temperature process from 80°C to -20°C . The output power was monitored by an optical power meter. The result shows that the output power decrease with decreasing temperature, and the sensor presents a quite good repeatability compared with the response of increasing temperature process, see Fig.7.

4. CONCLUSIONS

We have proposed a low cost and high sensitive temperature sensor based on a tapered optical fiber. The influence of the ambient refractive index and diameter of taper is simulated by RSOF BeamPROP. Reducing the taper waist diameter within a certain range can increase the sensitivity. A taper optical fiber about tens micrometers and several millimeters long was fabricated, and then a miniature temperature sensor which is made of the taper optical fiber coated by thermo-sensitive material is fabricated. The results of our experiments show a high sensitivity and a good repeatability. Due to the advantages of high sensitivity, good repeatability, simple fabrication and low cost, the proposed sensor can be found wide applications.

ACKNOWLEDGMENTS

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